COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

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Abstract. The Almgren–Pitts combinatorial argument is a recurring technical step in min-max constructions of embedded minimal hypersurfaces: a cover-refinement on the parameter interval, followed by a bookkeeping argument that turns local near-critical savings into a global supremum improvement. Each author re-derives it in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g), direct minimization over a fixed homology class may produce only degenerate configurations, and the relevant critical points are typically of saddle type. The resolution, following Birkhoff (1917) for closed geodesics and Almgren–Pitts for higher-dimensional varifolds, is a *min-max* construction; regularity of the output is Schoen–Simon [SS81].

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set* $\operatorname{Sing}\Sigma$ *has Hausdorff dimension at most $n-7$.*

The core technical result of Pitts's proof, used in every subsequent treatment [CDL03, DLT13, DLR18, DL16, MN14], is the Almgren–Pitts combinatorial argument, stated in the Colding–De Lellis formulation as follows.

Theorem 1.2 (Almgren–Pitts combinatorial Argument, [CDL03, Prop. 3.4])**.** *Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integral varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.*

The proof of Theorem 1.2 reduces to a one-parameter claim on the sweepout [DLT13, §3.2]:

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

The scalar formulation of the proof of the Claim, which is the precise theorem we formalize, reads as follows.

Theorem 1.3 (Extracted scalar version)**.** *Let X be a pseudo-metric space with an abstract paired-cover structure. Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$ and $m_0 > 0$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a* local witness*: an open*

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neighborhood $U_t \ni t$, a paired cover c_t in X, and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for every $s \in U_t$. Then there exists $f' : [0,1] \to \mathbb{R}$ with

$$\sup f' \;\leq\; m_0 - \frac{1}{4N}.$$

Translation from the De Lellis–Tasnady setting to Theorem 1.3: the boundary $\partial\Omega_t$ becomes a continuous $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U'_t) collapses to a single neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}$ collapses to a single continuous E_t; and the almost-minimizing hypothesis is no longer needed.

The Lean library actually proves a more general statement, of which Theorem 1.3 is a direct specialization: the parameter interval $[0,1]$ may be replaced by any compact, nonempty topological space K, and the specific pair $(1/N, 1/(4N))$ by arbitrary scalars (δ, ε) with $0 < \varepsilon \leq \delta$. The input is then a finite covering of the δ-super-level set of f carrying per-piece replacements with uniform saving ε.

Theorem 1.4 (Abstract core)**.** *Let K be a compact, nonempty topological space, $f : K \to \mathbb{R}$ a continuous energy with $m_0 = \sup f$, and let $0 < \varepsilon \leq \delta$. Suppose there is a finite family of pieces $\{p_l \subseteq K\}_{l \in \iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \geq \varepsilon\}$ with: (I) $f - E_l \geq s_l$ on p_l; (II) every $t \in K$ lies in at most two pieces; (III) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$. Then there exists $f' : K \to \mathbb{R}$ with*

$$\sup f' \;\leq\; m_0 - \varepsilon.$$

Theorem 1.3 follows by taking $K = [0,1]$, $\delta = 1/N$, $\varepsilon = 1/(4N)$, and running a 1D cover-refinement construction on the local witnesses to produce the family $\{p_l\}$ with two-fold overlap. Both theorems reappear in §2: Theorem 1.4 as the library's abstract API (Theorem 2.7, §2.2), and Theorem 1.3 as the one-parameter application (Theorem 2.8, §2.3).

Scope. This note formalizes the proof of the Claim (the combinatorial argument on $[0,1]$) as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the $(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships a single abstract theorem (§2.2) parametric in the sweepout space together with a one-parameter specialization to $[0,1]$ (§2.3); the formalization is machine-verified against `Mathlib`, declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space), `X` a pseudo-metric space (the ambient space whose covers witness replacements), and `f : K → ℝ` a continuous scalar energy.

In the proof of Theorem 1.2, the Lemma 3.1 of [DLT13] supplies, for each near-critical parameter t, an $\mathcal{H}^n$-disjoint pair of open sets $(U'_{1,t}, U'_{2,t})$ in M^{n+1} inside which a local replacement is performed. We abstract this pair as an instance of a class, stripped of its GMT content.

Definition 2.1 (`PairableCover` class)**.** [✓ `PairableCover`]

A `PairableCover X` instance declares, for each abstract `c : Cover`, a left region and a right region in X and a scalar diameter, subject to two axioms: the two regions are disjoint, and overlapping combined regions of diameter-comparable covers are ordered by inclusion.

The replacement family $\{\partial\Omega_{t,\tau}\}_{\tau\in[0,1]}$ at a near-critical t has its Hausdorff measure decreased by at least $1/(4N)$ inside U'_t. We keep only the scalar *replacement energy* $E : K \to \mathbb{R}$ and the uniform saving inequality; the geometric family is discarded.

Definition 2.2 (`LocalWitness` structure). [✓ `LocalWitness`]
A `LocalWitness K X f t` ε carries an open neighborhood $U \ni t$ in K, a cover `c : Cover`, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

The proof of the Claim works on the set $K \subset [0,1]$ of parameters at which the sweepout is within $1/N$ of the min-max value. We use the same convention.

Definition 2.3 (Near-critical set). [✓ `nearCritical`]
For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

At the end of the refinement the data is a family $\{\overline{J_l}\}$ covering K, with each J_l carrying a replacement whose energy saving is at least $1/(4N)$ and with two-fold overlap. We abstract this output: the near-critical threshold $1/N$ becomes a parameter δ and the saving floor $1/(4N)$ a parameter $\varepsilon \leq \delta$.

Definition 2.4 (`FiniteCoverWithWitnesses` structure). [✓ `FiniteCoverWithWitnesses`]
A `FiniteCoverWithWitnesses K f m₀` δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l\in\iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$.

Before the refinement, a Lebesgue-number argument on the compact set K produces a finite covering $\{I_i\}$ of open intervals, each centered at a $t_i \in K$. In our formalization the interval center c_i and the witness center w_i are recorded separately: grid points need not themselves lie in K.

Definition 2.5 (`InitialCover` structure). [✓ `InitialCover`]
Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries a finite family of open intervals $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ for $i = 1, \ldots, n$, each paired with a witness center $w_i \in$ NC and a local witness at w_i with saving $1/(4N)$, subject to (a) spacing: $c_i + r_i < c_{i+2} - r_{i+2}$ for all valid i; (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: NC $\subseteq \bigcup_i I_i$.[1]

The refinement is described as an iteration: each I_j with $j \geq 2$ is either kept whole or split into two halves. The state of this iteration after ℓ moves is the following.

Definition 2.6 (`PartialRefinement` structure). [✓ `PartialRefinement`]
Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic` ℓ carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k.

2.2. Public API: abstract core.

Theorem 2.7 (Abstract core). [✓ `exists_sup_reduction_of_cover`]
Input. *A compact, nonempty topological space K; a continuous $f : K \to \mathbb{R}$; $m_0 = \sup f$; scalars $0 < \varepsilon \leq \delta$; a* `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ *(Definition 2.4).*
Output. *There exists $f' : K \to \mathbb{R}$ with $\sup f' \leq m_0 - \varepsilon$.*

[1] Conditions (a) and (b) transcribe the spacing and subset conditions of the original proof.

2.3. Public API: one-parameter application.

Theorem 2.8 (One-parameter application). [✓ `exists_sup_reduction`]
Input. *A pseudo-metric space X with a* `PairableCover X` *instance (Definition 2.1); a continuous $f : \mathbf{I} \to \mathbb{R}$; $m_0 \in \mathbb{R}$ with $m_0 > 0$ and $m_0 = \sup f$; $N \in \mathbb{N}$ with $N > 0$; for every $t \in \mathrm{NC}$ a* `LocalWitness` *at t with saving $1/(4N)$ (Definition 2.2).*
Output. *There exists $f' : \mathbf{I} \to \mathbb{R}$ with $\sup f' \le m_0 - 1/(4N)$.*
Specializes Theorem 2.7 to $K = \mathbf{I}$, $\delta = 1/N$, $\varepsilon = 1/(4N)$.

2.4. Internal lemmas.

The declarations below are consumed by Theorem 2.8; proofs are sketched in §3.

Proposition 2.9 (Compactness of NC). [✓ `isCompact_nearCritical`, ✓ `nearCritical_nonempty`]
If K is compact and f continuous, NC *is compact. If additionally $m_0 = \sup f$ and $N > 0$,* NC *is nonempty.*

- ✓ `exists_initialCover` : existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- ✓ `exists_terminal_refinement` : existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- ✓ `not_three_overlap` : at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- ✓ `saving_bound_closure` : the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.
- ✓ `exists_refinement` : packages the 1D construction output as a `FiniteCoverWithWitnesses` with $(\delta, \varepsilon) = (1/N, 1/(4N))$ on $\mathbf{I}$; the input to Theorem 2.7.

Figure 1 shows the layered architecture.

3. Proof structure

We split the proof of Theorem 2.8 into three stages. Stage A builds an initial cover from the witness hypothesis. Stage B runs an induction producing a terminal partial refinement with injective σ. Stage C assembles the finite cover with witnesses by extending saving bounds to closures and deriving TwoFold. Theorem 2.7 closes.

3.1. Stage A: existence of the initial cover.

Proposition 3.1 (Existence of an initial cover). [✓ `exists_initialCover`]
Under the hypotheses of Theorem 2.8, an initial cover (Definition 2.5) exists.

Proof. Compactness of NC (Proposition 2.9) plus the open cover $\{U_t\}_{t \in \mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \dots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. Stage B: the refinement induction.

Starting from an initial cover, one builds pieces inductively.

Definition 3.2 (Base case). [✓ `step_zero`]
step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

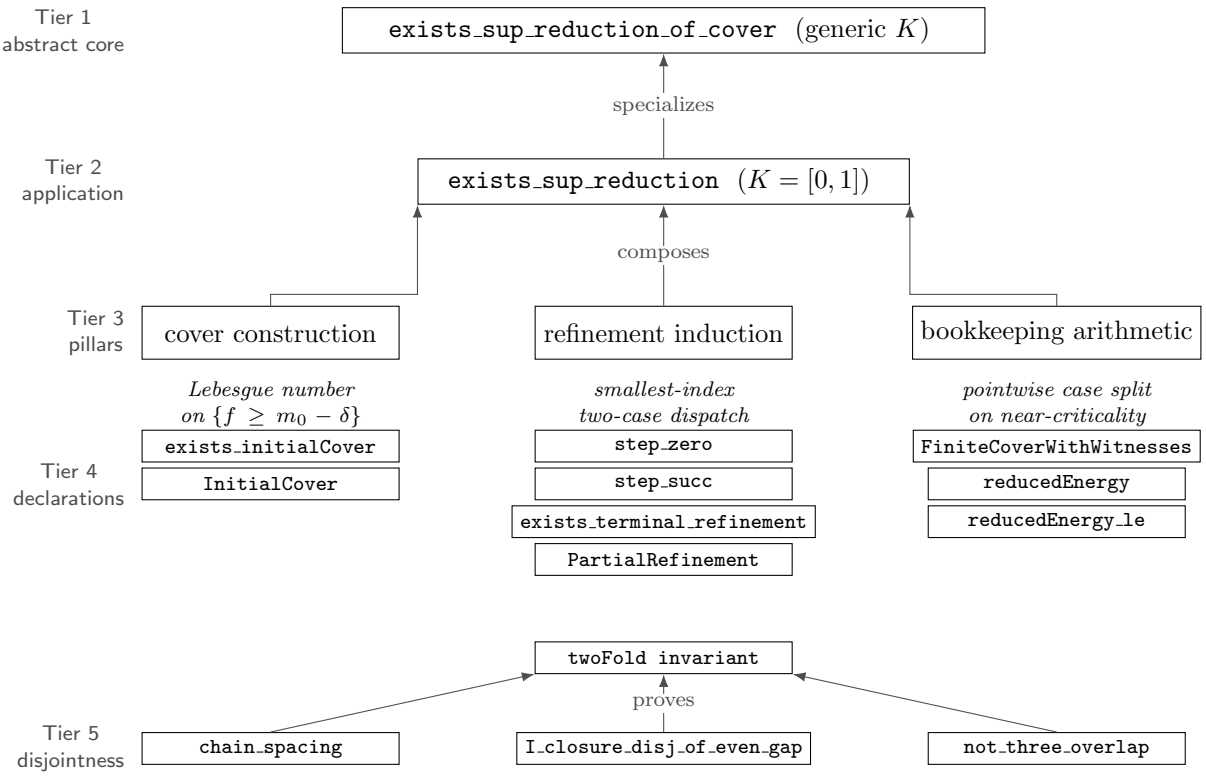

FIGURE 1. `CombArg` v0.1 architecture. The abstract core (Tier 1) is parametric in the sweepout space; the one-parameter application (Tier 2) instantiates $K = [0, 1]$ and composes three construction pillars (Tier 3). Each pillar's key declarations appear in Tier 4. The disjointness sub-module (Tier 5) supplies the two-fold overlap invariant consumed by bookkeeping.

Definition 3.3 (Inductive step). [✓ `step_succ_at`]

Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \ldots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.
(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[2]

Proposition 3.4 (Termination). [✓ `exists_terminal_refinement`]

Starting from step_zero(ic) *and iterating* step_succ_at, *one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.*

Proof. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in$ rem(pr) (chosen as the smallest such) produces pr′ with $i^* \notin$ rem(pr′) and rem(pr′) $\subseteq$ rem(pr)$\setminus\{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell+1) = i^*$ is fresh because $i^* \in$ rem while existing $\sigma(k)$ correspond to already-covered indices. □

[2] A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

3.3. **Stage C: assembly.** Stage C packages the refined covering as a `FiniteCoverWithWitnesses` (Definition 2.4), which the core theorem turns into the sup-reducing f'. We trace through [DLT13, §3.2, Step 2] in sequence, indicating at each step where the content lands in the formalization.

De Lellis and Tasnady begin by applying Lemma 3.1 to produce, at each near-critical parameter t, a replacement family $\{\partial\Omega_{i,t}\}$ with three properties: $\Omega_{i,t} = \Omega_t$ outside (a_i, b_i) and agrees with Ω_t outside U_i' inside (a_i, b_i); the global upper bound $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) + \frac{1}{4N}$; and, most importantly, the saving $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) - \frac{1}{2N}$ when $t \in (a_i + \delta, b_i - \delta)$. In our formalization this data is what a `LocalWitness` (Definition 2.2) carries: the scalar $E_{\sigma(k)} : K \to \mathbb{R}$ plays the role of $\mathcal{H}^n(\partial\Omega_{i,t})$, the neighborhood U_t plays (a_i, b_i), and the two separate bounds ("saving $-\frac{1}{2N}$ on the inner core" and "no-loss $+\frac{1}{4N}$ outside") collapse to a single uniform saving $\varepsilon = \frac{1}{4N}$ on U_t. Stage B has already handed us pieces p_l on which this witness saving is valid (Proposition 3.6), so by Stage C we have per-piece scalar savings $s_l \ge \varepsilon$ directly.

The next observation in the proof is that if $t \in (a_i, b_i) \cap (a_j, b_j)$, then $j = i + 1$, and moreover $\operatorname{dist}(U_i', U_{i+1}') > 0$. This is the two-fold overlap of the refined covering, and in our formalization it is the `twoFold` field of `FiniteCoverWithWitnesses`, derived from the parity-gap spacing by Proposition 3.5 below.

The modified sweepout $\{\partial\Omega_t'\}$ is then defined piecewise: $\Omega_t' = \Omega_t$ outside all J_i; $\Omega_t' = \Omega_{i,t}$ if t lies in a single J_i; and on $J_i \cap J_{i+1}$ the sweepout blends the two replacements, $\Omega_t' = [\Omega_t \setminus (U_i' \cup U_{i+1}')] \cup [\Omega_{i,t} \cap U_i'] \cup [\Omega_{i+1,t} \cap U_{i+1}']$. Stripped of the geometry this is additive scalar subtraction of savings over the pieces containing t: our reduced energy is

$$f'(t) \;=\; f(t) \;-\; \sum_{l\,:\,t \in \overline{J_l}} s_l,$$

and the two-fold invariant bounds the sum length by 2, matching the two-piece blend on overlaps.

The area accounting then splits on whether t is near-critical. For $t \notin K$, the point lies in at most two J_i's and each replacement can add at most $\frac{1}{4N}$, so

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) + \tfrac{1}{2N}.$$

For $t \in K$, the point lies in at least one inner core $(a_i + \delta, b_i - \delta)$, where the replacement saves $\ge \frac{1}{2N}$ in U_i', and in at most one other J_l where it costs $\le \frac{1}{4N}$ in U_l', for net

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) - \tfrac{1}{4N}.$$

In our formalization the same case analysis is the scalar argument for $f'(t) \le m_0 - \varepsilon$ inside the abstract core (the lemma `reducedEnergy_le` from §2.2): split on whether t lies in any piece of the cover, use the saving floor $s_l \ge \varepsilon$ in the positive branch and the covering condition $\{f \ge m_0 - \delta\} \subseteq \bigcup_l p_l$ in the negative branch. With $(\delta, \varepsilon) = (1/N, 1/(4N))$ this specializes to the arithmetic of the original proof; the abstract core reads off the same algebra for arbitrary (δ, ε) with $\varepsilon \le \delta$.

The rest of this subsection establishes the two propositions that Stage B leaves open: the saving bound extends from open J_k to $\overline{J_k}$ (continuity), and the two-fold invariant holds (a parity argument on the spacing condition (a)).

Proposition 3.5 (TwoFold via parity)**.** [✓ `terminal_twoFold`]

For a partial refinement pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \dots, \overline{J_L}$.*

Proof. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i-j| = 2(m+1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (✓ `chain_spacing`, ✓ `closure_disjoint_of_even_gap`).

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \dots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \dots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2:

either $c-a$ is even (and necessarily ≥ 2), or $c-a$ is odd and ≥ 3, in which case $b-a$ and $c-b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (✓ `not_three_overlap`). □

Proposition 3.6 (Closure extension of saving bound). [✓ `saving_bound_closure`]

Let pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \geq 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.*

Proof. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \geq 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.2 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\geq$-inequalities under limits. □

Lemma 3.7 (Assembly). [✓ `exists_refinement`]

A partial refinement pr *of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i yields a refinement of $(\mathbf{I}, X, f, m_0, N)$ in the sense of Definition 2.4, with $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and $E_k :=$ the replacement energy of the $\sigma(k)$-th witness.*

Proof. Invariant (I) is Proposition 3.6. Invariant (II) is Proposition 3.5. Invariant (III) holds by construction ($s_k = 1/(4N)$). Coverage of NC chains $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Theorem 2.8 now follows: chain Propositions 3.1 and 3.4, invoke Lemma 3.7, and apply Theorem 2.7.

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